Competitive Exam Practice 2: Number Systems, Representation, and Arithmetic Hardware

GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

Q1 *[GATE CS 2000 Q1.6, verified — GATEOverflow V3 p.351]* [**MCQ, 1 mark**] The number 43 in 2's complement representation is:

(A) 01010101

(B) 11010101

(C) 00101011

(D) 10101011

Q2 *[GATE CS 1994 Q2.7, verified — GATEOverflow V3 p.349]* [**NAT, 1 mark**] Consider n-bit (including the sign bit) 2's complement representation of integer numbers. The range of integer values N that can be represented is $-$ ________ $\leq N \leq$ ________ (write in terms of n).

Q3 *[GATE CS 1997 Q5.4, verified — GATEOverflow V3 p.350]* [**MCQ, 1 mark**] Given $\sqrt{(224)_r} = (13)_r$, the value of the radix r is:

(A) 10

(B) 8

(C) 5

(D) 6

Q4 *[GATE CS 1999 Q2.16, verified — GATEOverflow V3 p.288]* [**NAT, 1 mark**] The number of full adders and half adders required to add two 16-bit numbers in a ripple-carry adder is ________ full adders and ________ half adder(s).

Q5 *[GATE CS 2001 Q2.10, verified — GATEOverflow V3 p.351]* [**MCQ, 1 mark**] The 2's complement representation of $(-539)_{10}$ in hexadecimal is:

(A) ABE

(B) DBC

(C) DE5

(D) 9E7

Q6 *[GATE CS 2004 Q62, verified — GATEOverflow V3 p.289]* [**MCQ, 2 marks**] A 4-bit carry-lookahead adder, which adds two 4-bit numbers, is designed using AND, OR, NOT, NAND, NOR gates only. Assuming all inputs are available in both complemented and uncomplemented forms and the delay of each gate is one time unit, the overall propagation delay of the adder is (assume the carry network is implemented using two-level AND–OR logic):

(A) 4 time units

(B) 6 time units

(C) 10 time units

(D) 12 time units

Q7 *[GATE CS 1996 Q23, verified — web]* [**MCQ, 1 mark**] Booth's algorithm for integer multiplication gives the worst performance (i.e., the maximum number of addition/subtraction operations) when the multiplier pattern is:

(A) $101010\ldots1010$

(B) $100000\ldots0001$

(C) $111111\ldots1111$

(D) $011111\ldots1110$

Q8 *[Original, GATE-pattern]* [**NAT, 2 marks**] In the restoring division algorithm for two 8-bit unsigned integers (dividend and divisor loaded into a 16-bit AQ register pair with the divisor in a register M), how many total subtract/restore cycles (one per quotient bit) are performed to produce an 8-bit quotient?